Lab #10:- Plant Organs Part (1)

* Plant tissues are arranged or combined into organs that carry out life functions of the organisms.

Plant Organs

- Vegetative
  - roots
  - stems
  - leaves
- Reproductive
  - fruit
  - seeds
  - flowers

* The organ that bears the reproductive structures is called a cone, while in mosses is gametophores.

* The vegetative organs perform the vital functions, such as photosynthesis, while the reproductive organs are essential in reproduction.

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A- Vegetative Organs:-

1. The Root

* In vascular plants, the root is the non-leaf, non-nodes bearing parts of the plant's body that lies bellow the surface of the soil.

* Roots can be aerial or aerating (growing up above the ground or especially above water).

* Functions of Root :-

1- ~~Apsor~~ Absorption of water and inorganic nutrients.
2- Anchoring of the plant body to the ground, and supporting it.
3- Storage of food and nutrients.
4- Vegetative reproduction.

* In response to the concentration of nutrients, roots also synthesize cytokinin, which acts as a single as to how fast the shoots can grow.

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▶ Structure of the Root :-

I) A longitudinal section through the Root → divides into several regions :-

1* The Root Cap:-
- Formed by the lower portion of the aptical meristem differnt-iate or change into cells → arranged and form the Root Cap.

- Functions of the Root Cap:-
• Direct the growing of the ~~growing~~ root downward by sense of gravity.
• Secret smelly mucus like substance that lubricated and ~~protect~~ protects the root as it pass through the soil particles.
• Protect the meristem tissue.

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2* Apical Meristematic ⇒ Dividing cells that form tissues of other region.

3* Region of elongation:-

- Behind the apical meristem of the root.
- Elongate primarily by filling cell vacuoles with water.
- Growth of the root ⇒ Increase in :- number of apical meristem and size of individual cells of ~~this~~ this region.

4* Region of maturation:-

- Behind the region of elongation.
- The cells mature and begin to change their ~~structure~~ structure and function (differentiation).
- Produce Root Hairs → long extensions to increase the surface.

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II) Cross Section of Dicotyledonous Root:-

* Epidermis:-

- Single layer of cells on the outside to protect the inner tissue
- Doesn't have waterproof cuticle in the root.
- Root hair is out ward growth of the epidermal cells.

* The Cortex :-

- made up of parenchyma cells.
- Large cells → store water and food.
- facilitate the movement of water from the root hair cells on the outside of the plant to the xylem in the inside of the plant.

* The endodermis :-

- forms the innermost layer of the cortex.
- a layer of tightly-packed, modified parenchyma Known as the Casparian strip.
- This layer helps to regulate the flow of water from the cortex into the stele.

* The stele (vascular cylinder) :-

- Responsible for transporting of water and minerals.
- Made of :- pericycle, phelom and xylem.
- The pericycle ⇒ outer most layer, one or more rows of thin walled parenchyma cells and in close contact with xylem and phloem.

- Pericycle's function → forming of ~~lat~~ lateral root and lateral meristem to allow secondary growth (thickening) to occur.

* Xylem in dicot root arrange in an X arch shape.
* Phloem ~~in dicot root~~ tissues lay between the arms of xylem.

* Cambium seperates the xylem & phloem this where the secondary growth of these two occurs.

✱ Cross Section of Monocot Root :-

- The out most layer of the monocot root is surrounded by Epidermis & Ground layer differentiates into:- Cortex, Endodermis, pericycle and pith.

- The vascular bundle made up from 8 or more alternate bundles of xylem and phloem arranged in an alternate axis.

Differences between Dicot & Monocot Root. ⟸ الجدول ص ٩٠ من الشيوال حفظ

Good Luck^^

Lab # 10 : Plant Organs (Part 2)

2. Stem anatomy :-

* The main stem develops from the plumule of the embryo and the lateral branches develop from the buds.

* ~~Internodes~~ The leaves and lateral branches develop from the nodes and the region between them called internodes.

* Stems usually grow above the soil surface and towards the light from the sun.

* Stems → Herbaceous stems → leafy non-woody
  → Woody stems → harder than herbaceous stems.

* Functions of Stems :-

- Support for the plant as it holds leaves, flowers and fruit upright above the ground.
- Transport ~~and holding~~ of fluids between roots and shoots in the xylem and phloem.
- Storage of nutrients.
- Production of a new living tissue as it contains meristematic tissue.

I) Cross Section of Dicot Stem:-

1- Epiderms :-

- Single layer of living cells.
- The walls are thickened and covered with a thin waterproof layer called cutice.
- ~~Stomata~~ stomata with guard ~~cells~~ cells are found in ~~th~~ the epidermis.
- In some stems either unicellular or multicellular hair-like outgrowths appear from the epiderms.

* Functions of Epiderms :-

1. protects the underlying tissues.
2. The cuticle prevents the desiccation of inner tissues and thus prevents water loss.
3. The stomata allow gaseous exchange for the processes of respiration and photosynthesis.

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2- Hypodermis → beneath epiderms consist of collenchyma cells.

3- The Ground Tissue → differentiable into cortex, endodermis, pericycle, medullary rays and pith.

4- Vascular Cylinder or Stele :-

a- The Pericycle :

- made up of sclerenchyma cells (lignified, dead fiber cells).
- These cells have thick, woody walls and tapering ends.

Function of ~~Pericycl~~ Pericycle:-

* strengthens for the stem.
* Provides protection for the vascular bundles.

b- Vascular Bundles :-

* are situated in a ring on the inside of the pericycle of the plant (characteristic of distinguishing). It consists of Xylem, phloem, and cambium.

Xylem → inside
phloem → outside
cambium → between them, secondary thickening

c- Pith (Medulla) :-

- Occupies the large central part of the stem.
- Consists of thin-walled parenchyma cells with intercellular air spaces.
- between each vascular bundle is a band of parenchyma, the medullary rays, continuous with the cortex and the pith.

Functions :-

- The cells of the pith store water and starch.
- They allow for the exchange of gases through the intercellular air spaces.
- The medullary rays transport substances from the xylem and phloem to the inner and outer parts of the stem.

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II) Cross Section of Monocot Stem :-

1- Epidermis :- same as the one of the dicot stem and there's hypodermis beneath it.

2- Hypodermis :- internal to epidermis made from compactly arranged sclerenchyma without inter cellular spaces.

3- Ground Tissue :-

- Small, thick walled sclerenchyma on the inside of the epiderms.
- Intercellular spaces are found.
- No cortex or pith.

Functions of Ground Tissue :-

1- to store synthesized organic food such as starch.
2- Intercellular spaces allow the exchange of gases.

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4- Vascular bundles :-

- found scattered throughout the ground tissue.
- ~~occuring~~ & contain no cambium and consequently secondary thickening does not occur.

Parts of vascular bundles :-

ⓐ Sclerenchyma sheath → Thick walled sclerenchyma fibers surround the vascular bundle, which protect the vascular bundles and strengthen the stem.

ⓑ Xylem → Consists of vessels (→ Y-shape), tracheid, xylem parenchyma and limited xylem fibers. A large water cavity is present in the inner side of the protoxylem. Stores water and called water cavity.

ⓒ Phloem → lies outside the xylem in the vascular bundle, consists of sieve tubes, companion cells and phloem fibers, but lacks phloem ~~parenk~~ parenchyma.

(*) Outer phloem → protophloem $\xrightarrow{\text{func.}}$ met phloem lie inner portion.

Differences between Monocot & Dicot stem. ⇐ الجدول ص 95 من المانيوال حفظ

III) The Woody Dicot Stem :-

* A transverse section of a woody stem of 3-years old bass wood tree shows that it consists of :-

1- The bark :- (Cork, Cork cambium and phelloderm) → periderm, Cortex and phloem.

2- Vascular cambium :-

- adds new secondary xylem to the outside of the existing xylem.
- xylem area grows thicker and thicker.
- The new secondary phloem cells are added to the inner part of the ring of phloem.

(*) Cork Cambium → divide to form the protective cork.

3- Xylem :-

- Each year, the cambium produces a layer of secondary xylem and a layer of secondary phloem.
- The layer of secondary xylem each year is ~~most~~ much thicker than the layer of secondary phloem (because more of the derivatives mature toward the inside).

* annual ring
  - → spring rings → Xylem vessels made in the spring rings
  - → summer rings

4- pith → - In young stems, which is made of parenchyma, stores food.
- Disappears in old woody stems.

Good Luck ^^

Lab #10 :- Plant Organs (Part 3)

3- Leaf anatomy :-

- The leaf is the primary photosynthetic organ of the plant.
- Consists of a flattened portion, called the blade (a lamina).
- The blade is attached to the plant by a structure called the petiole.

- Leaves
  - → Simple leaves
  - → Compound leaves
  - → Doubly compound leaves

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▶ Internal structure of the Leaf :-

1- **Epidermis** :- outer layer covered with waxy cuticle which is impremeabl to liquid water vapor and in some it is thinner on lower epiderms.

(*) Functions of Epidermis :-

1- Prevent water loss by transpiration.
2- Regulation of gas exchange.
3- Secretion of metabolites.
4- Absorption of water in some species.

(*) Epidermis tissue cell types :-

1- Epidermal cells :- most numerous, largest, least specialized, and ~~more elongated in monocots~~ form the majority of the epidermis.

2- Epidermal Hair cell (trichomes).

3- Stomatal Complex Cells (guard and subsidiary cells):- function in regulation of gas exchange and water vaporization by opening and closing of stomatal pores that are surrounded by chloroplast containing guard cells at each side and two to four subsidiary cells.

* Stomata ~~more~~ more numerous over the lower epidermis.

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2* Mesophyll :- interior of the leaf between upper and lower epidermis made of parenchyma (ground tissue) or chlorenchyma.

* In ferns & most flowering plants, it consists of:-

1. ~~Pali~~ Palisade Layer :- one or two cells thick layer, these cells are columnar or vertically (cylindrical) elongated and regularly arranged in one to five rows.

* Chloroplasts in these cells are closed to their walls to take optimal advantage of light.

2. Spongy Layer :- irregular in shape and loosely arranged cells with large intercellular spaces for O_2 & CO_2 exchange connected to the pores of stomata through sub stomatal chambers.

* It contains fewer chloroplasts than Paliside Layer.

3* Veins :- the vascular tissue of the leaf located in the spongy layer. Its venation (pattern of veins) is parallel in monocots and interconnected network ~~s~~ in dicots.

* A vein is made of vascular bundles, which is wholly ~~surrounde~~ surrounded by ground tissue of lignin (sheath) to strengthen the rigidity of the leaf.

Differences between Monocot & Dicot Leaf ⟸ الجدول ص 102 من المانيوال حفظ

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B- Reproductive Organs :-

- Include flowers, seeds and fruit.

* Flowers :-

- The reproductive unit found in flowering plants (Angiosperms).

Flowers

- Perfect → have both male and female reproductive organs, the plant is said to be monoecious plant.
- Imperfect → have only male or ~~rep~~ female reproductive organs, the plant said to be dioecious plant
- Complete → have stamens, ~~pes~~ pistil, petales and sepals
- Incomplete → lack any part of complete flower

* A flower that lacks stamens is pistillate or female, while one that lacks ~~pa~~ pistils is said to be staminate or male.

* Functions of flowers :-

1- Effect reproduction by providing a mechanism for the union of sperm with eggs.

2- Used by humans to beautify their environments and as objects of romance, ritual, religion, medicine and ~~sourse~~ source of food.

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Floral Parts :-

* Basically, each flower consists of a floral axis contains :-

① Essential organs of reproduction → stamens and pistils.

② Accessory organs (vegetative) → perianth (~~s~~ sepals & petals) that attracts pollinating insects and protects the essential organs.

* A stereotypical flower consists of 4 kinds of structures attached to the tip through a short stalk, each of these kinds of parts in a ~~whole~~ whorle on the receptacle.

* The 4 ~~whole~~ main whorls (starting from the base of the flower)

a. vegetative part (perianth) :-

1) Calyx → the outer green whorl consisting of units called sapales which enclose to the flower in the bud stage, they may be absent or ~~p~~ prominent and petal like in some species.

2) Corolla → the next whrole toward the apex, composed of thin, soft and colored units called petals that attract insects to help in pollination.

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b. Reproductive Parts :-

1- Stamen :- the male reproductive organ consists of 2 parts:-

a* Anther → 2 pollen sacs, contains pollen grains.

b* Filament → Stalk like structure that support another.

2- Pistil :- the female reproductive organ consists of 3 parts:-

a* The Ovary → encloses the ovules, or potential seeds.

by carpel (simple pistil : one carpel
Compound pistil :- more than one carpel)

b* Stigma → the apex.

c* the Style → between ovary & stigma.

Good Luck ^^

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